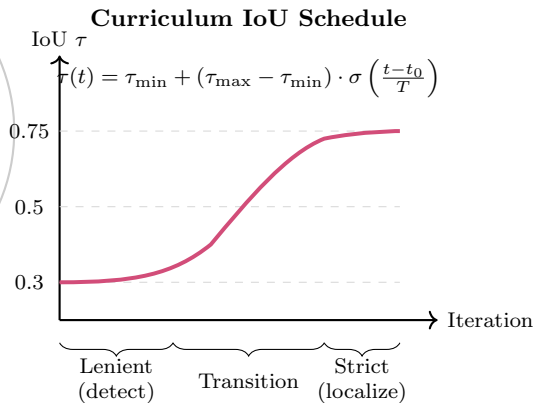
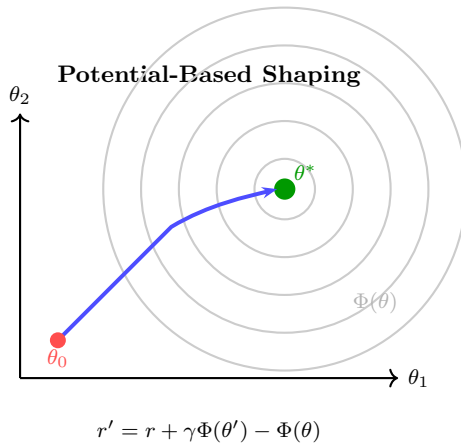
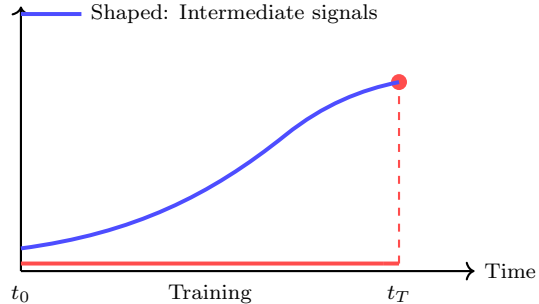
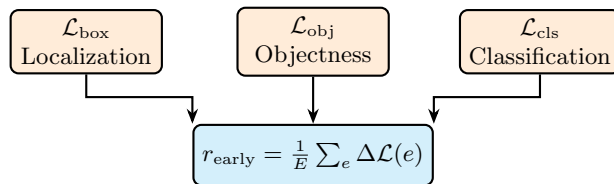


Sparse vs Shaped Rewards
 Sparse: r_{mAP}^* at t_T only
 Shaped: Intermediate signals



Intermediate Training Signals



- Benefits:
- Faster feedback
 - Guides exploration
 - Reduces variance